Simulation Description: `TestClocksCompetition`

MultiCellularClocks Project

March 28, 2026

1 Mathematical model

This document summarizes the mathematical model used by `TestClocksCompetition`. The node-based mechanical framework follows the overdamped centre-based formulation implemented in Chaste [2], and the logarithmic repulsive interaction is consistent with the Meineke-style spring law in the overlap regime [1].

1.1 State variables

Let $N = 16$ cells be indexed by $i \in \{1, \dots, N\}$. Each cell has:

- position $\mathbf{x}_i(t) \in \mathbb{R}^2$,
- radius r_i,
- drag coefficient $\gamma_i(t) > 0$,
- circadian value $c_i(t)$ stored in cell data.

For each run, cells are assigned to two phenotype sets:

- *non-cycling* set $\mathcal{N}$,
- *cycling* set $\mathcal{C}$,

with $|\mathcal{N}| + |\mathcal{C}| = N$. The current experimental setup uses three group compositions:

- All Non-Cycling: $|\mathcal{N}| = N$, $|\mathcal{C}| = 0$,
- All Cycling: $|\mathcal{N}| = 0$, $|\mathcal{C}| = N$,
- Half Non-Cycling / Half Cycling: $|\mathcal{N}| = |\mathcal{C}| = N/2$, with identity assigned by random shuffle at initialization.

1.2 Neighbour interactions

For each interacting pair (i, j), define

$$\mathbf{d}_{ij} = \mathbf{x}_j - \mathbf{x}_i, \qquad d_{ij} = \|\mathbf{d}_{ij}\|, \qquad \hat{\mathbf{d}}_{ij} = \frac{\mathbf{d}_{ij}}{d_{ij}}, \tag{1}$$

$$\ell_{ij} = r_i + r_j, \qquad \delta_{ij} = d_{ij} - \ell_{ij}. \tag{2}$$

A pairwise force is applied only in overlap (repulsion) regime $d_{ij} < \ell_{ij}$:

$$\mathbf{F}_{ij} = p\, \hat{\mathbf{d}}_{ij}\, \ell_{ij}\, \log\left(1 + \frac{\delta_{ij}}{\ell_{ij}}\right), \tag{3}$$

where $p > 0$ is the `RepulsionParameter` (set to 1.0 in this test). Equation (3) is evaluated using the geometric quantities from Eq. (2).

The cell-interaction force contribution is drag-weighted asymmetrically:

$$\mathbf{f}_{i\leftarrow j} = \frac{1}{\gamma_i}\mathbf{F}_{ij}, \tag{4}$$

$$\mathbf{f}_{j\leftarrow i} = -\frac{1}{\gamma_j}\mathbf{F}_{ij}. \tag{5}$$

This force-based centre model is discussed in detail in [3].

Thus higher drag reduces the applied mechanical contribution.

1.3 Circadian signal and drag law

A global circadian signal is defined by

$$c(t) = \sin\left(\frac{2\pi t}{T} + \phi\right), \qquad T = 24,\ \phi = 0. \tag{6}$$

This periodic form is the canonical phenomenological choice for circadian phase forcing [4].

Phenotype-dependent rules are piecewise:

- If $i \in \mathcal{N}$:

$$c_i(t) = 2.0, \qquad \gamma_i(t) = 1.0. \tag{7}$$

- If $i \in \mathcal{C}$:

$$c_i(t) = c(t), \tag{8}$$

  and with threshold $\theta = 0$:

$$\gamma_i(t) = \begin{cases} 0.1, & c(t) > \theta, \\ 10.0, & c(t) \le \theta. \end{cases} \tag{9}$$

Hence Eq. (9) creates alternating high-mobility and low-mobility phases synchronized to Eq. (6).

Therefore cycling cells alternate between low-drag and high-drag phases, while non-cycling cells keep constant drag. The three simulation groups differ only by composition $(|\mathcal{N}|, |\mathcal{C}|)$; all other mechanical and cell-cycle parameters are held fixed.

1.4 Equation of motion (overdamped node dynamics)

For each node i, the net applied force is

$$\mathbf{F}_i^{\text{net}}(t) = \sum_{j\in\mathcal{N}(i)} \mathbf{f}_{i\leftarrow j}, \tag{10}$$

where $\mathcal{N}(i)$ is the interaction-neighbour set from the node-based mesh. By construction, Eq. (10) aggregates pair terms from Eq. (5).

In overdamped form, velocity is proportional to net force,

$$\dot{\mathbf{x}}_i(t) \propto \mathbf{F}_i^{\text{net}}(t), \tag{11}$$

and the simulation integrates with time step $\Delta t = 0.01$ to final time $T_{\text{end}} = 240$.

1.5 Initial and experimental conditions

- Initial positions: square lattice, spacing 1.0.
- Interaction radius in neighbour search: 1.5.
- Simulation groups: All Non-Cycling, All Cycling, and Half Non-Cycling / Half Cycling.
- Replicates: 10 runs per group with seeds $0, 1, \ldots, 9$.
- Cell-cycle model: Bernoulli trial with contact inhibition, using `QuiescentVolumeFraction` = 0.75, `EquilibriumVolume` = 1.0, `DivisionProbability` = 0.02, `MinimumDivisionAge` = 4.0.

2 Parameter summary

Quantity	Value
Number of cells N	16
Replicates	10
Time step Δt	0.01
Final time T_{end}	240
Circadian period T	24
Circadian phase ϕ	0
Threshold θ	0
Repulsion parameter p	1.0
Cycling-cell drag (high circadian phase)	0.1
Cycling-cell drag (low circadian phase)	10.0
Non-cycling drag	1.0

3 Interpretation

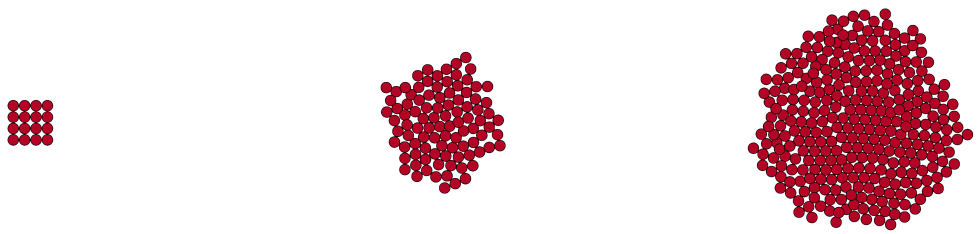
The model compares population-scale behaviour across three phenotype compositions under shared mechanics. The cycling phenotype experiences periodic drag switching driven by the circadian signal, while the non-cycling phenotype has constant drag. This isolates the effect of composition (all non-cycling, all cycling, or mixed) on spatial dynamics and cell-number trajectories under the same pairwise repulsion law.

4 Figure placeholders for simulation groups

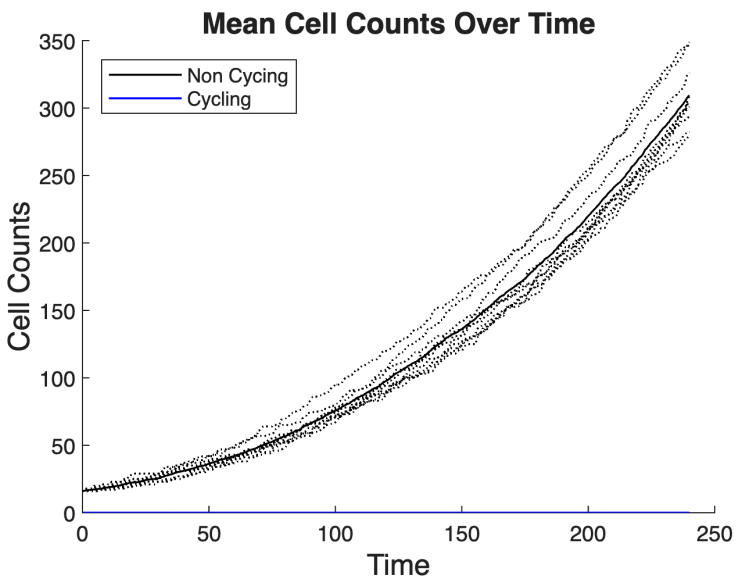
The following figures are placeholders for each simulation group. For each group, three snapshot time points are shown ($t = 0$, $t = 120$, and $t = 240$) together with a cell-number-over-time panel.

Simulation groups:

- All Non-Cycling,
- All Cycling,
- Half Non-Cycling / Half Cycling.



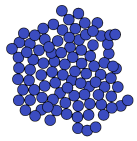
(a) $t = 0$ days (b) $t = 5$ days (c) $t = 10$ days



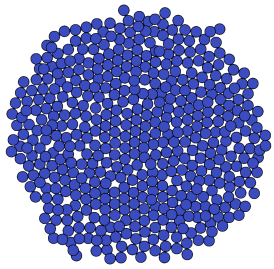
(d) Cell number over time

Figure 1: All Non-Cycling simulation group: three snapshot timesteps and cell number over time.

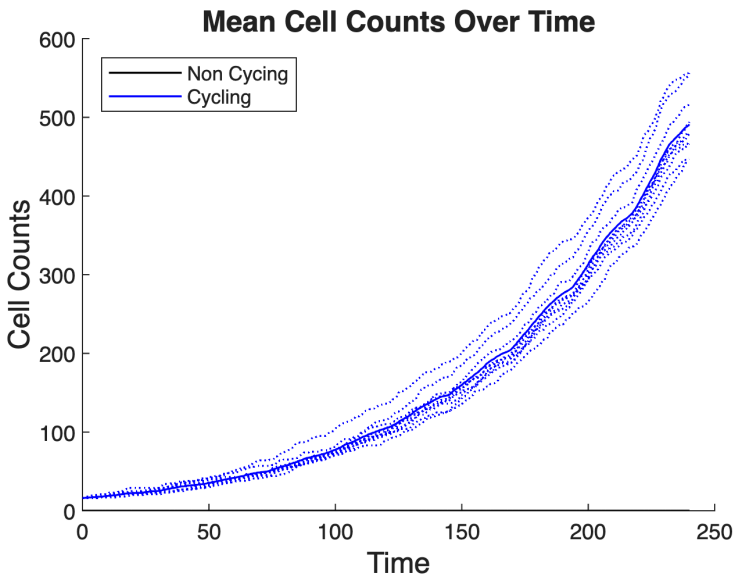
(a) $t = 0$ days



(b) $t = 5$ days

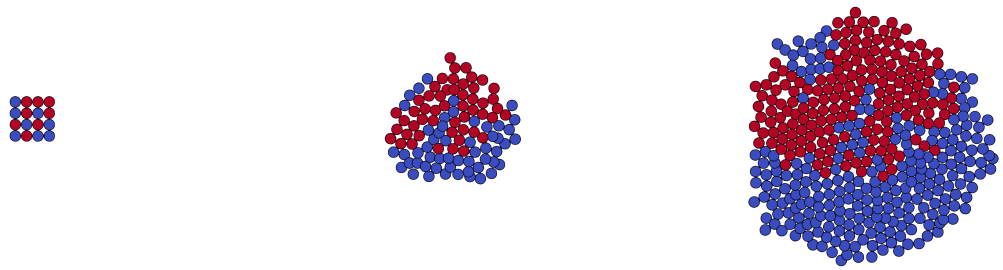


(c) $t = 10$ days

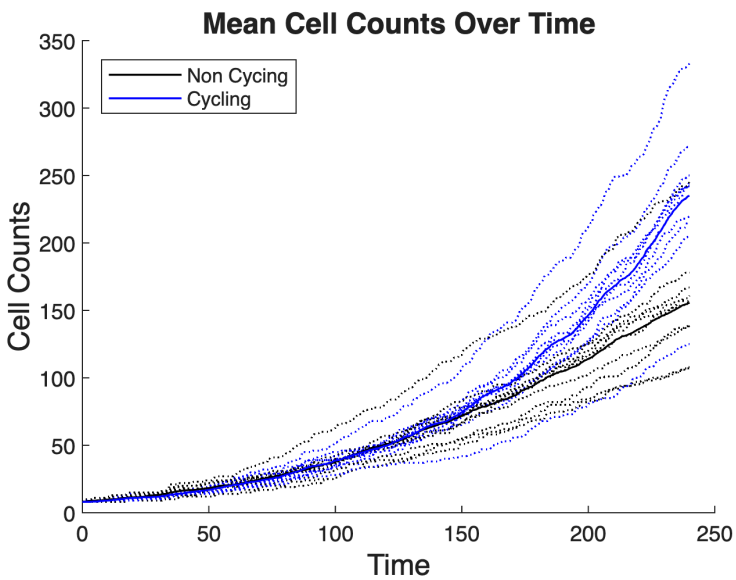


(d) Cell number over time

Figure 2: All Cycling simulation group: three snapshot timesteps and cell number over time.



(a) $t = 0$ days (b) $t = 5$ days (c) $t = 10$ days



(d) Cell number over time

Figure 3: Half Non-Cycling / Half Cycling simulation group: three snapshot timesteps and cell number over time.

References

[1] F. A. Meineke, C. S. Potten, and M. Loeffler. Cell migration and organization in the intestinal crypt using a lattice-free model. *Cell Proliferation*, 34(4):253–266, 2001.

[2] G. R. Mirams, C. J. Arthurs, M. O. Bernabeu, R. Bordas, J. Cooper, A. Corrias, Y. Davit, S.-J. Dunn, A. G. Fletcher, D. G. Harvey, M. E. Marsh, J. M. Osborne, P. Pathmanathan, J. Pitt-Francis, J. Southern, N. Zemzemi, and D. J. Gavaghan. Chaste: an open source c++ library for computational physiology and biology. *PLOS Computational Biology*, 9(3):e1002970, 2013.

[3] J. M. Osborne, A. G. Fletcher, J. M. Pitt-Francis, P. K. Maini, and D. J. Gavaghan. Comparing individual-based approaches to modelling the self-organization of multicellular tissues. *PLOS Computational Biology*, 13(2):e1005387, 2017.

[4] Joseph S. Takahashi. Transcriptional architecture of the mammalian circadian clock. *Nature Reviews Genetics*, 18(3):164–179, 2017.